# Software as a Service
## towards a service science

Franco Zambonelli

April 2014



# Outline

- Software as a service
  - Product vs services
  - Cloud computing
- Everything is a service
  - Examples of non traditional services
  - Humans, Robots, Mobility
- Towards a service science
  - Service science as a computing science
  - Service science as a management and economic science
- What do we need to know?
  - Complex systems, social systems, multiagent systems

## Software as a Service

- Words, Excel, Win
  - Are software products
  - The idea is to build and sell a product
- Google, Google Docs, Maps
  - Are software services
  - The idea is to sell the service, not a product
- The Software as a Service paradigm
  - Access on need
  - Pay per use

3

## Cloud Computing

- Move stuff out of the desktop
  - Access on-demand to software and infrastructures
  - Pay per use (or pay indirectly, e.g., advertisement)
- Software as a service, but also
  - Platform as a service → even the OS becomes a service
  - Infrastructure as a service → hardware resources accesses on need
- You no longer need to "buy" but
  - You rent as you need
  - You always get the best (up-to-date)
  - You do not need to perform maintenance (24/7 availability)

4

## But is this about ICT only?

- NO!
  - The shift towards "services" is pervasive in modern economics and society!
- ICT is only making all of this possible
  - By making it possible to create infrastructure for the dynamic provisioning of services
  - Supporting service interactions and service findings
  - Service-oriented architectures and middleware for the real-world

5

## Example: the IKEA Case

- IKEA
  - Does IKEA really sell furniture?
  - To some extents, it sells a life style ("a service for better living")
  - But it indeed sells a service
- It does not build furniture
  - It orchestrate a complex service of furniture provisioning
  - It ensures quality in such orchestration
  - We add value (by assembling)
  - IKEA adds value by coordinating

6

## Example: The Apple Ecosystem

- Apple does no longer sells PCs or phones
  - It sells gates to a whole ecosystem of services
- The higher price of iPhones and MacBooks is not intrinsic in the objects
  - It has nothing to do with production prices
  - Is the price for accessing to the Apple ecosystem

7

## Example: from Cars to Mobility

- Big car industry are shifting from
  - Selling cars to
  - Selling mobility services
- The "car" per se is no longer the key focus ("look at what wonderful car I can sell you")
  - The focus is on "mobility" (look at how well you can move by buying the car)
  - The car become simply a mean to access mobility services

8

## Cars and Added Value

- So is the added value in the car?
- No:
  - A car with no streets and no parkings has NO value
  - The value is in the network of streets, parkings, and other cars
  - In the way they interact
  - The car adds value to streets and viceversa → cocreation

9

## Traditional Economy: Good-dominant logic

- The focus is on "goods"
- Value in exchange for goods (or work for producing goods)
- The added value is in the goods themselves
- Networking is limited to bilateral actions

- This is changing very rapidly
  - Sad to see very few economists perceiving that

10

## Traditional Economy: Service-dominant logic

- The focus is on "services"
- Value in applying knowledge to put services at work
- The added value is always in a process of "value co-creation"
- Goods (operant resources) are simply a means to exchange services (operand resources)
- Networking has to account for the whole network of value cocreation and its dynamics

11

## Key Principles of SD-logic

**Table 2** Foundational premises of S-D logic

| Premise number | Foundational premise |
|---|---|
| FP1 | Service is the fundamental basis of exchange. |
| FP2 | Indirect exchange masks the fundamental basis of exchange. |
| FP3 | Goods are a distribution mechanism for service provision. |
| FP4 | Operant resources are the fundamental source of competitive advantage. |
| FP5 | All economies are service economies. |
| FP6 | The customer is always a co-creator of value. |
| FP7 | The enterprise can not deliver value, but only offer value propositions. |
| FP8 | A service-centered view is inherently customer oriented and relational. |
| FP9 | All social and economic actors are resource integrators. |
| FP10 | Value is always uniquely and phenomenologically determined by the beneficiary. |

12

## Service-dominant Economy vs Good-dominant one

| GD | SD |
|---|---|
| Working | Applying knowledge |
| Selling a good | Selling a service possibly where goods are only "access and distribution means" |
| Added value of goods in transformation | Added value of services in integration and value co-creation |
| Value chain | Value networks |
| Value in exchange | Value in use |
| Complex production networks | Complex social networks and multiagent networks |
| Management as work coordination | Management as complex dynamics understanding and control |
| Engineering manufacturing | Engineering interactions |

## And so…

- Being ready for the next service economy implies
  - Understanding the dynamics of complex systems
  - Understanding complex social networks
  - Understanding multiagent systems

14